



Multimodal fusion diagnosis of the Alzheimer's disease via lightweight CNN-LSTM model using magnetic resonance imaging (MRI)

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ABSTRACT

Alzheimer's disease is categorized as a primary neurodegenerative ailment that mostly affects individuals in the elderly age and those reaching later stages of life. The recognition of this conditions may be ascribed to the persistent cognitive processes that display similarities to the state of anxiety. The timely identification of Alzheimer's disease (AD) and its precursor, mild cognitive impairment (MCI), occurring potentially up to 8 years prior to the manifestation of dementia symptoms, has great significance. The incorporation of Neuroimaging methods are now indispensable in the assessment and timely identification of Alzheimer's disease (AD). The use of artificial intelligence in the medical domain has become prevalent, and deep learning has emerged as a promising approach for computer-assisted Alzheimer's disease diagnosis using magnetic resonance imaging (MRI). This paper presents a unique, efficient, and precise multimodal fusion deep learning architecture that utilizes Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM). In this study, we used a well-established dataset called Alzheimer's Disease Neuroimaging Initiative-1 (ADNI-1) as the gold-standard dataset for detecting Alzheimer's disease (AD). ADNI-1 included 3D characteristics, which were transformed into 2D features and then employed in our innovative model. Subsequently, the innovative model was subjected to training and evaluation using the aforementioned dataset. Based on the empirical findings, the newly introduced model achieved a remarkable accuracy of 92.3 %, surpassing the accuracies achieved by the current leading baseline models while using fewer parameters and requiring less processing complexity.

1. Introduction

Alzheimer's disease (AD) is a neurodegenerative condition characterized by cognitive decline and behavioral abnormalities. According to clinical symptoms, the classification of individuals may be categorized into Mild Cognitive Impairment (MCI), Normal Control (NC), and Alzheimer's disease (AD) [1,2]. The prevalence of dementia has seen a significant rise due to the advancing age of the worldwide population. The average remaining lifetime after diagnosis typically ranges from three to nine years, as shown by previous research [3–5]. Based on [6], it has been determined that Alzheimer's disease (AD), along with other forms of dementia, has a global impact on around 50 million individuals. According to a study conducted by [7], there has been a significant rise in the worldwide prevalence of AD. The number of individuals affected by this condition has escalated to 43.8 million in 2016, representing a

substantial increase of 117 % in comparison to the 20.3 million recorded in 1990. The aforementioned figure is projected to see a threefold increase, reaching a total of 152 million by the year 2050. Globally, the incidence of dementia is projected to occur at a rate of one new case every three seconds [8].

Advanced medical imaging techniques, such as positron emission tomography (PET) and magnetic resonance imaging (MRI), have greatly improved the accuracy of diagnosing a number of diseases and disorders. Magnetic resonance imaging (MRI) has become more common in clinical settings due to its affordability, ease of acquisition, and higher resolution, which allows for thorough three-dimensional imaging. As a result, MRI data has been used in a number of scientific investigations [9]. Brain morphometric analysis employing structural magnetic resonance imaging (MRI) has shown effective in detecting morphological differences between AD and non-AD individuals and in tracking the

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progression of moderate cognitive impairment (MCI). Therefore, a growing number of researchers are focusing on Alzheimer's disease diagnosis by MRI image analysis of patients [10,11].

Machine learning has greatly improved Alzheimer's disease (AD) diagnosis by training models in the last few decades, to predict classifications of unidentified cases by extracting relevant features from AD data. Areas of interest (ROIs) such as the temporal lobe, bilateral hippocampus, entorhinal cortex volume, grey matter (GM), white matter (WM), and structural thickness abnormalities are often manually extracted using this method [12]. Experts have also developed multimodal multitask learning [13], which they use to diagnose diseases using standard algorithms like L1-Support Vector Machine (SVM) and Random Forest (RF). Combining feature learning with tailored detection approaches has enhanced functional connectivity studies [14]. In order to generate novel classification features, these solutions often use dimensionality reduction and preprocessing techniques. The underlying premise of several ROI identification methods in magnetic resonance imaging (MRI) is that changes in brain tissue lead to cognitive deterioration [15]. Aspects at the ROI level highlight structural changes [16], whereas known coordination points allow for the assessment of anomalies relevant to Alzheimer's diagnosis [17]. Research shows that using many parameters improves diagnosis accuracy, as seen in chest X-ray evaluations [18]. Along with exploring the use of the whole brain image as a major feature, conventional methods have been examined, including Histogram of Gradients (HoG), Haar-like features, and Local Binary Patterns. In order to extract useful information from high-dimensional magnetic resonance imaging (MRI) scans of the brain, dimensionality reduction methods are required, most notably Principal Component Analysis (PCA) [19]. One study employed random forest classification to diagnose Alzheimer's disease after extracting features using 3D-HOG and principal component analysis [20]. A distinct body of work developed a complex model for the multi-aspect and traditional method-based detection of serious diseases in medical images [21]. The use of a 2D/3D automated encoder might also make dimensionality reduction and feature extraction easier. In order to reliably diagnose Alzheimer's disease, high-dimensional brain MRI data has been transformed into low-dimensional representations using a convolutional auto-encoder model, and then an SVM algorithm was used [22].

Deep learning techniques have recently come to the fore in computer vision and are finding widespread use in the detection of Alzheimer's disease (AD), as a means of enhancing ensemble diagnostic tools. Numerous diagnostic methods, mostly using multi-layer convolutional architectures, have been supported by a single deep learning model, as shown by [23,24]. The current state-of-the-art deep learning approaches to diagnosing Alzheimer's disease using whole volume MRI images are complex and time-consuming to train. The need for simpler methods that use effective networks to improve diagnostic accuracy is highlighted by the fact that complexity does not always correlate with efficacy. Typically, deep learning approaches to diagnosing Alzheimer's disease include a two-step procedure: first, certain areas of the image are used to extract features, and second, these features are combined for categorization. However, most existing methods [25,26] mainly focus on obtaining image data from disease-related brain regions, ignoring the importance of spatial interactions and connections between other brain regions. The topological structure and spatial connectivity of the brain's discriminative regions as shown in magnetic resonance imaging (MRI) may vary between diseases. Therefore, local texture information, topological data across brain areas, and spatial correlations are all necessary for an accurate diagnosis of Alzheimer's disease. Importantly, structural changes in MRIs are often linked to clinical features, and it is vital to understand that brain shrinkage affects specific regions. A significant challenge in utilizing deep learning algorithms for MRI-based AD diagnosis is identifying critical brain areas severely impacted by AD or MCI and enhancing certain features within those regions.

Our study aims to address practical challenges and recognized problems by developing an innovative module that blends global

similarity and local variety in medical imaging. With a simplified model that delivers better performance, we want to achieve complete identification. To effectively diagnose Alzheimer's disease (AD) using MRI data, we provide a direct multimodal fusion approach that incorporates deep learning components. Enhancing Alzheimer's disease detection techniques is the primary objective. The main points of our research are outlined below:

- *An innovative and precise model.* In this study, we present a novel multimodal fusion deep learning approach that combines Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) architectures. This model is characterized by its lightweight nature, quick processing speed, and high accuracy, as it has shown superior performance compared to existing state-of-the-art baselines.
- *3D to 2D feature transformation.* The 3D feature vectors have been transformed into 2D data in order to enhance their compatibility with the deep learning model proposed in this study, as well as in pre-trained state-of-the-art deep networks. The aforementioned transformation serves as a practical illustration of the process of converting from a three-dimensional representation to a two-dimensional representation.
- *Light-weighted variant.* The proposed model exhibited the capacity to accurately categorize a provided test image, achieving an average processing time of 1.7 ms. The processing time shown a significant improvement in comparison to the lightweight state-of-the-art methods.
- *A comprehensive set of SOTA methods.* A thorough assortment of total of 21 classifiers were used in our study to serve as the comparative benchmarks for evaluating the effectiveness of the innovative model provided.

The subsequent sections of the paper are organized in the following manner: [Section 2](#) under "Related work", provides a comprehensive overview of the existing literature and research in the field. [Section 3](#) provides a detailed description of the materials and methods employed in the present study. [Section 4](#) describes the experimental findings and subsequent discussion. [Section 5](#) includes a conclusion of our work and offers suggestions for future research.

2. Related work

In this part, we will present a concise overview of the earlier strategies based on traditional and deep learning techniques used for automatic Alzheimer's disease (AD) detection utilizing magnetic resonance imaging (MRI).

2.1. Overview of conventional approaches utilized in the diagnosis of Alzheimer's disease (AD)

There has been a significant amount of attention given to feature selection methods aimed at aiding the diagnosis of Alzheimer's disease (AD) through the analysis of individual brain magnetic resonance imaging (MRI) scans. A series of research investigations are presented to employ manually created features taken from MRI data in conjunction with various models. Zhang et al. [27] presented a multi-task feature selection strategy, the purpose of which is to locate subsets of features derived from a number of different modalities. An inter-modality feature selection technique known as IMTFS was introduced by Liu et al. [28] for the goal of analyzing supportive inter-modality data. Stefan et al. [29] employed a variety of data sources in their research in order to gather useful MRI biomarkers. These biomarkers were fed into a procedure called linear discriminant analysis. Cao et al. [30] carried out, a particular extraction of subset characteristics and Regions-of-Interest (ROIs) was carried out. These extracted features were included into a multi-task learning model with the intention of making the diagnosis of Alzheimer's disease (AD) easier. The AD dataset reported by Cilia et al.

[31] included 174 individuals' samples of handwriting. Data on handwriting is collected when people with Alzheimer's disease and a healthy control group complete handwriting tasks. The anatomy of the hippocampus was represented using spherical harmonics by Gerardin et al. [32]. They also developed a system for categorizing patients for more accurate diagnosis. The initial study on the detection of Alzheimer's disease (AD) was done by Shi et al. [33], who developed a nonlinear feature engineering module and employed it in combination with the support vector machine (SVM).

2.2. Overview of machine learning approaches utilized in the diagnosis of Alzheimer's disease (AD)

Machine learning techniques often include the extraction of quantitative features from magnetic resonance imaging (MRI) data, followed by the application of traditional classifiers. The process of feature extraction has significant importance in the realm of machine learning methodologies. The use of highly discriminating traits has the potential to successfully train the classifier and efficiently differentiate across classes. Ahmed et al. [34] used a fusion technique to combine characteristics derived from the hippocampus and posterior cingulate cortex. These fused features were then utilized to train support vector machine (SVM) classifiers for the purpose of diagnosing Alzheimer's disease (AD) and mild cognitive impairment (MCI). Lei et al. [35] introduced a technique for selecting features from multi-modal brain networks. They used support vector machines for prediction purposes, which led to a notable enhancement of 3.76 % in classification accuracy when compared to the utilization of fMRI alone. Chitradevi et al. [36] used a range of optimization methodologies to disintegrate the brain into its constituent elements, including the hippocampus, white matter, and gray matter. Hett et al. [37] introduced an innovative grading approach that explores the potential of multidirectional texture maps generated by the use of 3D Gabor filters. The objective of this technique is to enhance the accuracy of Alzheimer's disease (AD) diagnosis. Nawaz et al. [38] introduced three distinct machine learning models and conducted a comparative analysis to determine the model that exhibits superior accuracy in the detection of Alzheimer's disease (AD). Plocharski et al. [39] conducted a classification analysis to distinguish between individuals with Alzheimer's disease (AD) and cognitively normal (CN) individuals. This classification was based on the sulcal surface characteristics, which were recovered by calculating a medial surface from sulcal meshes. Khatri et al. [40] developed a methodology that combined a filter and wrapper approach for feature selection in order to improve the accuracy of Alzheimer's disease (AD) diagnosis utilizing several biomarkers. This approach shown a considerable improvement in the diagnostic accuracy. Sørensen et al. [41] proposed employing the hippocampal characteristics seen in sMRI scans as a means to aid in the diagnosis of Alzheimer's disease (AD). This approach is justified by the well-established understanding that the hippocampus is notably impacted by the degenerative progression of AD.

Machine learning techniques often used in the diagnosis of Alzheimer's disease sometimes rely on manually designed features in order to develop classification models. Nevertheless, the use of features that are retrieved individually and then processed by classifiers may result in less than ideal diagnostic performance owing to possible variations across these features.

2.3. Overview of deep learning approaches utilized in the diagnosis of Alzheimer's disease (AD)

Deep learning has recently shown substantial breakthroughs in the field of computer vision, which has therefore provided a broad range of insights for the detection of Alzheimer's disease (AD). It is possible to acquire different dimensional characteristics from magnetic resonance imaging (MRI) data by making use of convolutional neural networks (CNNs) or other deep learning architectures. This eliminates the

requirement for labor-intensive manual feature extraction. The ability of enhanced features produced from magnetic resonance imaging (MRI) scans to capture complicated and non-linear correlations between these features and the disease state has the potential to improve the accuracy of Alzheimer's disease (AD) diagnosis. Billones et al. [42] incorporated a dropout mechanism following each pooling layer of the VGG-16 model in order to determine which people were affected by Alzheimer's disease, mild cognitive impairment, or neither. The accuracy that was reached as a result was 91.85 %. Almadhoun et al. [43] used the transmission of Xception learning architectures to process the full brain image in their research. Subsequently, a CNN with detachable convolutional layers was constructed in order to extract and learn the key features of the image, which ultimately made it possible to classify images of the brain. Folego et al. [44] proposed an innovative methodology for the identification of many categories of biomarkers associated with Alzheimer's disease (AD). The researchers used an end-to-end deep three-dimensional convolutional neural network (3D CNN) in their methodology, whereby the entire image volume was used as input. Basaia et al. [9] proposed the development of a specialized three-dimensional convolutional neural network (3D CNN) tailored for the analysis of magnetic resonance imaging (MRI) data. The primary aim of their study was to effectively distinguish between Alzheimer's disease (AD), clinically confirmed moderate cognitive impairment (c-MCI), and suspected mild cognitive impairment (s-MCI) without relying on any pre-existing feature design. Lei et al. [45] introduced a convolutional network model that leverages longitudinal data from several time points to accurately identify individuals afflicted with Alzheimer's disease (AD). Lian et al. [16] proposed the use of a hierarchical fully convolutional network for the purpose of automatically extracting multi-scale feature representations from structural magnetic resonance imaging (sMRI) data encompassing the complete brain. The primary objective of this approach was to enhance the diagnostic process of Alzheimer's disease (AD). Zhu et al. [46] devised a tailored network architecture specifically designed for the identification and diagnosis of Alzheimer's disease (AD). The proposed network integrates interpretable feature learning with dynamic graph learning inside a graph convolutional network (GCN) architecture, yielding a diagnostic tool that is both resilient and reliable. Hedayati et al. [47] used an unsupervised convolutional autoencoder for the purpose of feature extraction from 3D images that underwent registration using several templates. Subsequently, the researchers acquired classification results by using an additional convolutional neural network (CNN). The technique exhibited a degree of accuracy of 95 % on ADNI dataset. Chen et al. [48] used a hybrid approach including iterative sparse and deep learning methodologies to get representations of critical brain regions. These representations are then employed in the diagnostic process of Alzheimer's disease (AD). Gopinath et al. [49] introduced a novel graph convolutional network that is designed to handle surface-valued. The network is capable of producing subject-based classification and regression results. Su et al. [50] introduced the use of feature distributions derived from brain images at many places for the purpose of binary classification. This was achieved via the implementation of domain adaptation approaches. Despite the promising outcomes shown by these models, the majority of previous methodologies have been constructed with intricate components that pose challenges in terms of complexity and understanding.

3. Material and method

This section provides a detailed description of the following subsections: (i) Dataset description, (ii) 3D to 2D image transformation, (iii) software stack of the implemented end-to-end software, (iv) suggested new model, (v) baseline models, and (vi) training of the proposed model.

3.1. Dataset description

The data used in the preparation of this paper were acquired from the

Alzheimer's disease Neuroimaging Initiative-1 (ADNI-1) database, a publicly accessible dataset designed for the examination of Alzheimer's disease diagnosis. The Alzheimer's Disease Neuroimaging Initiative (ADNI) has been conducting research to investigate the viability of using a combination of serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, as well as clinical and neuropsychological assessment, to monitor the progression of mild cognitive impairment (MCI) and early Alzheimer's disease (AD). The Alzheimer's disease Neuroimaging Initiative-1 (ADNI-1) public dataset [51] was utilized for both training and testing in this study. The dataset comprises 3D-structural MRI scans of individuals diagnosed with Alzheimer's disease (AD), moderate cognitive impairment (MCI), and normal controls (NC). In addition, the Alzheimer's Disease Neuroimaging Initiative (ADNI) consistently monitor and revise the dataset pertaining to individuals who have been identified as being in a pre-diagnosed state. Finally, the experimental dataset consisted of a total of 505 T1-weighted 3D-structural MRI scans obtained from different participants, including 135 patients with Alzheimer's disease (AD), 215 patients with mild cognitive impairment (MCI), and 155 cognitively normal controls (NC). All scans were sourced from the Alzheimer's disease Neuroimaging Initiative (ADNI). In this, the Clinical Dementia Rating (CDR) and Mini-Mental State Examination (MMSE) are used to assess the cognitive and clinical status of the individuals. To facilitate the integration of the three-dimensional attributes included in the ADNI dataset with the proposed model and state-of-the-art pre-trained deep neural networks (DNNs), a conversion process was used. This process involved the transformation of the three-dimensional features into two-dimensional representations via the utilization of transfer learning techniques. This facilitated the representation of three-dimensional attributes as two-dimensional elements. The process of transformation will be further examined in detail in [section 3.3](#). In order to mitigate the risk of data leaking, as discussed in research on AD diagnosis [52], the data is partitioned prior to undergoing preprocessing. The presence of identical scans of the same person in both the training and testing sets has been shown to artificially enhance the performance of diagnosis [53,54]. In order to achieve balanced data partitioning with comparable scale distributions across the training, validation, and testing sets, various clinical criteria are taken into account. These criteria primarily involve the utilization of the Mini-Mental State Examination (MMSE), global clinical dementia rating (CDR), and clinical dementia rating scale sum of boxes (CDR-SOB) scores, along with demographic information.

The construction of a dataset included a feature dimension resizing operation, resulting in a total of 505 images with dimensions of $448 \times 448 \times 3$ required by the proposed models and state-of-the-art pre-trained deep neural networks. Dimension of $448 \times 448 \times 3$ was chosen in order to enhance the classification performance and augment the information available inside the network. This resizing operation increases the dimensions of the images to 448×448 , surpassing the typical input size of $224 \times 224 \times 3$. The demographic and clinical information of the participants is shown in [Table 1](#). The dataset was divided into a training set comprising 70 %, and a test set comprising 30

% of the data. The specifics of the dataset created for the proposed study are outlined in [Table 2](#).

3.2. Proposed method

The model under consideration is a novel architectural design that incorporates Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) layers. [Fig. 1](#) illustrates the comprehensive framework of the innovative model developed for the diagnosis of Alzheimer's disease (AD) using magnetic resonance imaging (MRI) specifically pertains to the three-class classification of Alzheimer's disease (AD) that is Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer's disease (AD) subjects. The architecture has a total of 18 layers, starting with an Input layer, designed to process 2D RGB images with dimensions of 448×448 pixels containing three color channels. The provided images depict the two-dimensional representations of the extracted features derived from the samples inside the ADNI-1 dataset. The process of conversion has been thoroughly elucidated in detail in [sub-section 3.3](#). Subsequently, the input was sent through the first step consisted of a convolutional layer with 64 filters, allowing convolution operations to be carried out on the data. Following that, a task of hyper parameter optimization was carried out in order to determine the suitable activation function for the dense layer. The purpose of this investigation was to evaluate the most appropriate activation function amongst many different activation functions that are often used in neural networks. These included several variety of the Rectified Linear Unit (ReLU) [79], as well as softmax, hyperbolic tangent (tanh), exponential Linear Unit (eLU) [80], and a particular type of ReLU known as LReLU [81], which was employed as the activation function in the dense layer. The Leaky Rectified Linear Unit (LReLU) enhanced training efficiency over conventional activation functions. The LReLU activation function was selected due of its low processing cost and absence of saturation, which avoids vanishing gradients. No node ought to remain idle, hence the LReLU activation function solves the dying ReLU issue. The Leaky Rectified Linear Unit (LReLU) keeps negative values by giving a small slope instead of zero. The usage of the Batch normalizing layer, which was originally introduced by [82] requires the normalizing of the output of each activation. According to Salimans et al. [83] the normalizing procedure entails determining the average and standard deviation of the outputs over all of the samples that make up the mini-batch. After the Convolutional layer comes the Batch normalizing layer, which is utilized to achieve the required level of normalizing after the Convolutional layer. Afterward, the method referred to as Max Pooling was used, which is grounded on the principle of sub-sampling, and it was utilized to reduce the dimensions of the input images. The use of the maximum function across a pooling window [84,85] facilitated the achievement of this task. The use of Max Pooling resulted in many positive repercussions such as a decrease in training time for the Convolutional Neural Network (CNN) model, a reduction in the hardware resources needed to store tensors owing to a smaller image size. Lastly it has desirable implications on the enhancement [86] in the general efficiency of the system. Input width and height were decreased in half using a Max Pooling layer with a pool size of (2×2) . As a consequence of this, a significant fraction of the activations that originated from the preceding layer were eventually removed or ignored. Because of the expanded size and complexity [87] of deep neural networks, one of the most serious challenges [88] that has arisen is over-fitting. A Dropout [89] layer was implemented in order to solve the problem of over-fitting that is often seen in deep neural networks. A dropout rate of 0.4 was

Table 1

Demographic and clinical information (mean $\pm$ standard deviation) of the studied ADNI subjects. AD: Alzheimer's disease; MCI: mild cognitive impairment; NC: normal control; M: male; F: female; MMSE: Mini-Mental State Examination; CDR: Clinical Dementia Rating.

Diagnosis	Subjects	Gender (M/F)	Age	MMSE	Education (year)	CDR
NC	155	84/71	76.21 $\pm$ 7.21	24.08 $\pm$ 1.40	15.2 $\pm$ 2.4	0.7 $\pm$ 0.2
AD	135	77/58	74.54 $\pm$ 6.89	23.54 $\pm$ 2.34	15.4 $\pm$ 2.9	0.6 $\pm$ 0.16
MCI	215	124/91	75.32 $\pm$ 5.64	25.11 $\pm$ 6.31	15.5 $\pm$ 2.5	0.01 $\pm$ 0.12

Table 2

The details of the dataset constructed for the proposed study.

Set	Number of samples	Shape of sample
Training	378 (75 %)	$448 \times 448 \times 3$
Test	127 (25 %)	$448 \times 448 \times 3$

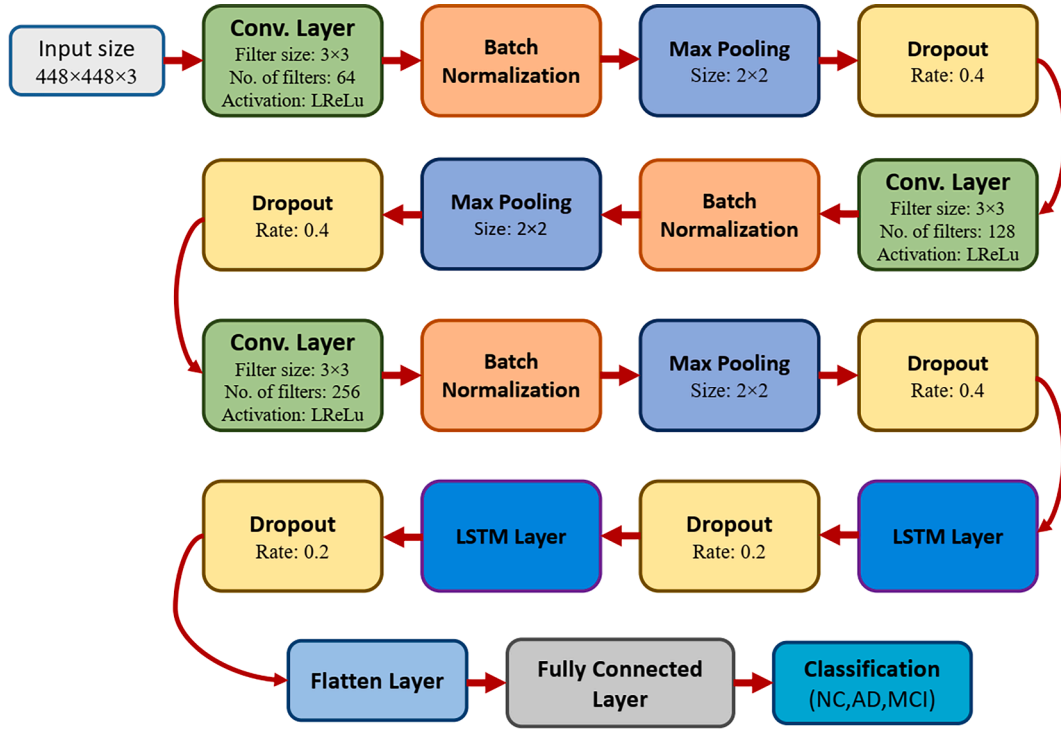


Fig. 1. An overview of the architectural design of the proposed innovative model, which consists of a total of 14 layers.

configured for the Dropout layer in the configuration settings. During the training process, dropout is used as a technique to randomly eliminate units, also known as neurons, from the neural network. The purpose of this approach is to mitigate the occurrence of co-adaptation. To mitigate the issue of over-fitting, several Dropout layers were strategically included at different stages. Furthermore, Max Pooling layers also serve the purpose of mitigating over-fitting. Then, we utilized two further series of convolutional layers, each of which was followed by Batch Normalization, Max Pooling with a pool size of (2×2) , and Dropout with a dropout rate of 0.4. Convolutional layers were used, with 128 and 256 sizes, respectively, for each of the two series' filters. Afterwards, we utilized a total of two Long Short-Term Memory (LSTM) [90] layers, each of which was then followed by a dropout layer with a dropout rate of 0.2. The Long Short-Term Memory (LSTM) model is a distinct variant of the recurrent neural network (RNN) [91] architecture. In this work, it is used to solve difficult computational issues because of its capability to efficiently incorporate previous knowledge with the present task. Long Short-Term Memory (LSTM) models were especially devised to address the challenge of long-term dependencies in recurrent neural networks (RNNs) that arise due to the vanishing gradient problem. Long Short-Term Memory (LSTM) models are equipped with specialized memory units that possess the ability to keep and disseminate information through extended sequences. The recent progress in LSTM design has led to notable enhancements in its ability to effectively capture and represent intricate temporal connections. The use of this characteristic enables Long Short-Term Memory (LSTM) models to assess whole sequences of data without the need to consider each individual data point in isolation. The LSTM network uses a cell state to store and retain long-term states, which encompasses the input, forget, and output gates. In our proposed model, the CNN architecture is used for the purpose of extracting deep features, whereas the LSTM model is typically utilized for detection tasks by leveraging the features that have been recuperated. The output generated by the Convolutional Neural Network (CNN) is then inputted into a Long Short-Term Memory (LSTM) network, which has been specifically developed to effectively capture and model long-term dependencies present within the input data. The Long Short-Term Memory (LSTM) network acquires the ability to

represent the sequential pattern of features that have been extracted by the Convolutional Neural Network (CNN). LSTM layers are crucial for capturing temporal dependencies in sequential data, which is particularly important in our application where the relationship between frames or time steps can substantially influence the outcome. The sequential nature of the data is not inherently addressed by Convolutional Neural Network (CNN) layers, despite their ability to effectively extract spatial features from images. Our model can gain an expanded understanding of the underlying patterns by incorporating LSTM layers, which can leverage both spatial and temporal information. This combination improves the model's capacity to make precise predictions by preserving critical contextual information over time. In comparison to models that exclusively utilize CNNs, our experimental results indicate a substantial enhancement in accuracy when this hybrid architecture is implemented. Afterward, we employed a Flattening layer as an intermediary between the previous network and the Fully Connected layers, to transform the input into a vector [92] before feeding it into the Fully Connected layer. The proposed model's output layer, comprised of a fully connected layer with three units and the softmax activation function, classified the resulting image into three classes: (1) *Cognitively Normal (NC)/healthy*, (2) *Alzheimer's disease (AD)*, and (3) *mild cognitive impairment (MCI)*.

The proposed model uses Xavier initialization [93] in order to achieve efficient and consistent convergence. The Xavier initialization technique effectively manages and ensures the correct initialization of weights. Both the Convolutional (Conv) and dense layers' bias vectors were given starting values of zero when the layers were initialized. The proposed model's convolution layers have a filter size of (3×3) , representing the convolution window dimensions computed during hyperparameter optimization. The Convolutional layers of the proposed model used a stride of (1×1) , indicating that one step was taken along both the height and width dimensions.

Hyper-parameters refer to the parameters inside a deep neural network (DNN) model that have a direct impact on the learning process. These parameters are determined by empirical methods. To optimize the utilized hyper-parameters, a comprehensive evaluation was conducted on a diverse set of regularly utilized values for each hyperparameter.

This analysis aimed to identify the optimal combination of hyperparameters. Table 3 shows the employed hyperparameter values. Ray Tune [94] is a very robust and adaptable framework for hyperparameter tuning, which may be effectively used in conjunction with diverse deep learning frameworks, such as PyTorch. The approach offered demonstrates scalability and efficiency in the context of hyperparameter optimization. The study included a comprehensive hyperparameter optimization employment via Ray Tune, including a total of 355 trials. The purpose of the optimization employment was to pick the accuracy gained for the validation set. The proposed approach uses the Adam as its optimization technique, using a learning rate of 0.01. Regularization improves deep neural network generalization to reduce test error. The utilization of *weight decay* was implemented in this study. This regularization method sums squared weights and multiplies by a hyperparameter. To prevent over-fitting, a value of $1 \times e^{-5}$ was utilized for *weight decay* alongside the Dropout and Max Pooling layers that were already employed. The assessment of the model's predictive capability is quantified by the use of the loss function, a fundamental component within the domain of deep learning. The primary aim of this procedure is to achieve a state of steady convergence for the model, which requires the minimization of the loss function. The architecture we propose incorporates the focal Tversky loss (FTL) [95] as a loss function. The goal of this loss function aims to minimize the occurrence of false-negative predictions and separate data instances that offer significant challenges. The loss function may be defined as:

$$\text{LossFunction}_{\text{FTL}} = (1 - \frac{[I_{\text{GT}} \cap I_{\text{PRE}}]}{[I_{\text{GT}} \cap I_{\text{PRE}}] + \alpha(I_{\text{GT}} - I_{\text{PRE}}) + \beta(I_{\text{PRE}} - I_{\text{GT}})})^{\frac{1}{\gamma}} \quad (3)$$

The parameter γ is constrained within the interval [1,3], and its purpose is to accentuate the impact of a localized gastric cancer region on the overall loss function. The hyper parameters α and β may be tuned in order to mitigate the issue of class imbalance. The variable I_{GT} denotes the ground truth image, whereas I_{PRE} represents the matching predicted image. The symbol $\cap$ is used to represent the intersection between the ground truth and predicted image.

3.3. 3D MRI image volume transformation into a 2D image

To incorporate the three-dimensional (3D) attributes present in the ADNI (1) dataset into the proposed model and state-of-the-art pre-trained deep neural networks (DNNs), a conversion was performed utilizing transfer learning methods. This conversion transformed the 3D characteristics into two-dimensional (2D) features. The use of the conversion approach in the previous studies conducted by [55–57] served as an additional motivating element for the creation of the present conversion procedure. Fig. 2 presents a graphical depiction of the procedure involved in converting 3-dimensional attributes into 2-dimensional images.

Initially, the individual segments of the three-dimensional medical image were subjected to the Transform network, which facilitates the conversion of three-dimensional images into a two-dimensional fused

image by merging all the slices. In order to address the calibration of input 3D patches, we propose the extension of the SE block [58] as a solution. The Squeeze-and-Excitation block is a computational module that may be constructed using a transformation function F_{tr} . This function translates an input X to feature mappings Z , as $Z = [z_1, z_2, \dots, z_k]$

The model that was used in our research successfully condenses the global spatial information included inside each slice via the utilization of the calculation of the average value found in each slice which can be computed as:

$$y_k = F_{sq}(z_k) = \frac{1}{P \times Q} \sum_{i=1}^P \sum_{j=1}^Q z_k(i, j) \quad (1)$$

The variable y_k represents the overall embedded information inside the specific slice denoted as $k.P \times Q$ represents the spatial dimensions of Z .

The second procedure, which is referred to as “excitation,” is utilized to capture dependencies across slices by using a sigmoid (σ) and ReLU (δ) activation function, respectively to normalize the slice weight between 0 and 1. Therefore, we have:

$$c = F_{ex}(y, M) = \sigma(M_2 \delta(M_1 y)) \quad (2)$$

The weight matrices M_1 and M_2 represent the fully connected layers with a reduction ratio of r .

Finally, the scalar c_k and input slice z_k are multiplied together in order to generate the calibrated three-dimensional patch, R' . A single slice is sent through a shallow Convolutional Neural Network (CNN) model, consisting of five convolutional layers, in order to get a singular scalar value for each slice. In our model, the congestion layer is represented by a block that is made up of five different convolutional layers. The kernel size of the first three layers is equal to 3×3 , while the kernel size of the latter two layers is equal to 1×1 . Through the use of nonlinear dimensionality reduction, this setup makes it possible to represent calibrated three-dimensional slices in two dimensions.

Consequently, every two-dimensional patch generates a three-dimensional representation that resembles a section of the magnetic resonance volume, consisting of a sequence of n consecutive slices. This paradigm enables the integration of certain 3D data without the need for extensive processing power and memory resources often associated with 3D convolutional neural networks (CNNs).

3.4. Overview of the training details of the proposed model

For training of the proposed model, the images used from ADNI databases were accessible in either PNG or JPG format. All of the datasets—the train dataset, the validation dataset, and the test dataset—had their resolution equalized to 448 x 448. The whole dataset is partitioned into two subsets for the purpose of conducting experiments on them: the training dataset (comprising 70 % of the data), and the test dataset (comprising 30 % of the data). In order to guarantee consistent and accurate results from an experiment, the k -fold cross-validation process is used. A 5-fold cross-validation approach was employed for robust experimental outcomes. Based on hyper-parameter optimization, k was set at 5. The training set—70 % of the dataset—was divided into 5 equal folds. First fold was validation set, then 4 folds were training folds. The aforementioned method was iterated five times in order to fully use the whole dataset for both training and validation objectives. The training procedure incorporates a supervisory mechanism to enhance the identification of significant features, particularly in diagnosis tasks. Furthermore, the acceleration of the training process is achieved by using batch normalization to normalize the feature matrix. In this particular model, the optimizer Adam was selected for its efficient computing capabilities. The model consists of 100 neurons, employs the Leaky Rectified Linear Unit (LReLU) activation function, and maintains a regularization rate of $1 \times e^{-5}$. Additionally, the model undergoes a total of 300 epochs. A loss function was included into the process of training

Table 3

Evaluated values for each hyper parameter values during hyper parameter optimization task.

Hyperparameter	Selected value
Conv Layers Dropout rate	0.4
Pooling layer	Max-pooling
Optimization method	Adam
Activation function	Leaky ReLU
Regularization rate	$1 \times e^{-5}$
Filter size in Conv layers	3×3
Number of folds(k)	5
Learning rate	$1 \times e^{-3}$
Fully connected layer Dropout rate	0.2

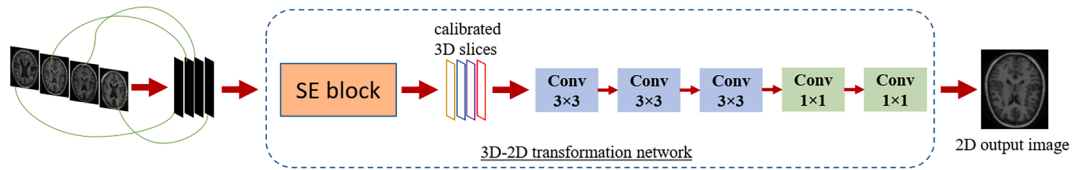


Fig. 2. 3D to 2D transformation network.

the model, and its purpose was to measure the difference that existed between the predictions generated by the model and the actual true values. The reduction in the loss function, which occurs when the model is being trained, is an indication of how well the data have been fitted. When the learning process is exhausted and the model is no longer taking in any new information, the early stopping technique is used for ending the monitoring conditions and the number of epochs that the approach observes before terminating are configured to determine the early stopping strategy.

4. Experimental results and discussion

This section outlines a set of experiments conducted to validate the effectiveness of the proposed architecture, in comparison to traditional machine learning and deep learning models. The studies were performed using a publicly accessible database. The dataset used in this study is derived from the Alzheimer's Disease Neuroimaging Initiative-1 (ADNI-1). The proposed research relies on experiment hardware since certain results depend on time, which is fully dependent on hardware capabilities. Notice that all tests were reproduced 15 times, and the mean of the numerical data from these ten trials was the final result.

4.1. Data pre-processing

In order to effectively train a model, it is necessary to get clean sample of data, as emphasized by Ren et al. [59], clean samples refer to balanced and noise-free data. After analyzing neuroimaging modalities, it's important to understand how various researchers employ them in deep learning architectures to diagnose Alzheimer's disease. Neuroimaging modalities undergo a variety of preprocessing steps to prepare data for research. An autonomous diagnostic system needs preprocessing to succeed. The system's ultimate success is heavily dependent on its execution of these steps. There are several neuroimaging data preprocessing tools nowadays. Some examples include FSL [65], FreeSurfer [60], DARTEL [9], and SPM [61,62]. In order to mitigate the potential negative impact of extraneous information in brain MR images on the efficacy of our classification model, we opted to use the FSL and FreeSurfer image analysis package for cortical reconstruction and volumetric segmentation.

These approaches include skull stripping, intensity normalization, tissue segmentation, registration, motion correction, and Gradwarp. Outputs corresponding to these processes are shown in Fig. 3.

- **Skull stripping:** in which the skull is removed from the normalized image, is an important preprocessing step.
- **Intensity normalization:** The purpose of the intensity normalization procedure is to eliminate fluctuations in intensity. To get a mean intensity of 110 for white matter, the intensities of all voxels are correspondingly scaled. The N4 approach [63] was used to address the problem of non-uniform tissue intensities. Additionally, an affine alignment technique was employed to match the structural magnetic resonance imaging (sMRI) with the MNI152 space, which serves as the normalized template. The minimum-maximum normalization technique is used to perform intensity normalization within the range of [0, 1].
- **Tissue segmentation:** The tissue segmentation method used by [9,64] is designed to partition a brain image into distinct segments, such as white matter (WM), grey matter (GM), and cerebrospinal fluid (CSF), in order to calculate their respective volumes.
- **Registration:** The FSL bet tool [65] is used for the purpose of extracting non-brain tissue from whole head imaging. Additionally, the FSL flirt tool [66,67] is employed to execute affine registration in MNI152 space, therefore aligning the image with the standardized template.
- **Motion correction:** In instances when there are several sources of volume, this procedure will effectively address and rectify any slight movements that may occur between them, ultimately amalgamating them into a cohesive average.
- **Gradwarp:** The proposed method aims to address the issue of geometry distortion resulting from gradient non-linearity, by implementing a gradwarp [68] correction technique.

4.2. Data augmentation

Data augmentation methods are used in order to expand the quantity of the existing data and mitigate the issue of over-fitting. In order to supplement the CNN training dataset, we conducted an investigation into several data augmentation methods, including flipping, rotation, and translation. This phenomenon may be more accurately denoted as class balance or oversampling, as opposed to augmentation. In rotation-based data augmentation, input images undergo rotation at several angles, including 90°, 180°, and 270°. On the other hand, flip augmentation involves reflecting the images horizontally and vertically. Several researches in the relevant literature have also examined image translations; yet, this may lead to inaccurate outcomes in the segmentation procedure. The significance of data augmentation lies in its ability to generate synthetic data from an existing dataset. In this work, we

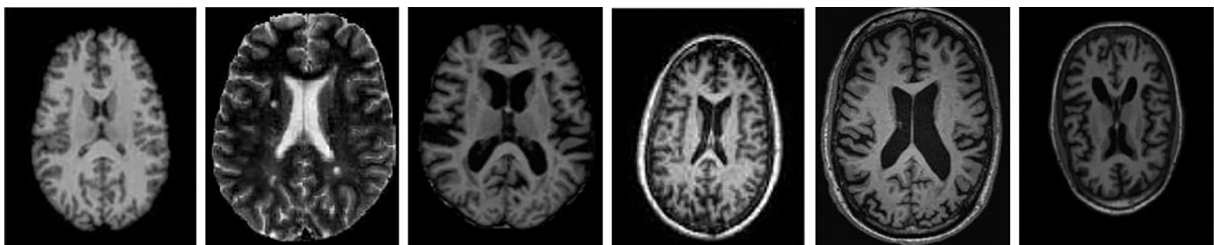


Fig. 3. The image supplied depicts the sequential transformation of a subset of the dataset sample after various pre-processing procedures conducted by FSL, FreeSurfer, arranged in a left-to-right manner.

employed the synthetic minority oversampling technique (SMOTE) as a data augmentation strategy to address the issue of class imbalance. The proposed methodology employs a linear transformation of feature vectors to provide additional data points by leveraging existing data points from the minority class.

The primary goal of data augmentation improving our model's robustness, especially in cases when our training data is limited. The 3D MRI scans in our dataset were captured from three different angles, but by converting them to 2D images, we can simplify the model input and improve computation speed. Rotating, flipping, and scaling are data augmentation methods that may be used to 2D MRI images to increase the diversity of the dataset. This augmentation not only alleviates overfitting but also improves the model's capacity to generalize across various scenarios, hence enhancing the precision and dependability of our decision-making process. The findings of this data augmentation experiment can be observed in Fig. 4.

4.3. SOTA models

In this study, we used a diverse set of SOTA classifiers, namely 26 classifiers, as the fundamental models for our analysis. The proposed innovative approach used a variety of classic machine learning and deep neural network models as its baselines. In order to achieve this objective, a selection of 12 commonly used classical machine learning models were utilized, namely (1) Naïve Bayes, (2) AdaBoost, (3) Random Forest, (4) Decision tree, (5) Independent component analysis (ICA), (6) Logistic regression, (7) K-nearest neighbors algorithm (KNN), (8) Learning vector quantization (LVQ), (9) Linear SVM, (10) XGBoost, (11) K-means, and (12) Genetic algorithm. In this study, conventional machine learning models were combined with a group of 14 recently developed pre-trained deep neural networks, namely (1) ResNet152, (2) Inception-v4, (3) MobileNet-V2, (4) Xception, (5) GoogLeNet, (6) SqueezeNet, (7) VGGNet, (8) AlexNet, (9) DenseNet201, (10) NaSNet, (11) SENet, (12) EfficientNet-B7, (13) YOLO-V5, and (14) RetinaNet, explicitly referred to as DNNs. In order to make the most of these networks, the technique of transfer learning was employed. This approach aims to improve a learner's performance in a certain domain by providing them with information and abilities from a domain that is connected to the original

domain. In the present study, information was transmitted without training pre-trained model layers. Inspired from [96], the famous ImageNet [97] dataset, with over 15 million tagged images in 22,000 categories, was used to train these models. Therefore, ImageNet weights were utilized to represent data transit. In order to streamline the transfer learning process, the exclusion of the classification layers was deemed necessary, given these layers were originally tailored for particular classification tasks. In the present study, the fully connected layer was trainable, unlike pre-trained model layers. In the proposed research, the classification aim was achieved by progressively including a fully connected layer with three units and using the softmax activation function into each pre-trained model. The aforementioned procedure persisted until the suggested model attained its ultimate state. Utilizing the weights of the layers that were produced from the ImageNet dataset allowed for a considerable reduction in the number of trainable parameters that were included in the pre-trained models. Fig. 5 depicts the steps used to build the pre-trained models from which the baseline models were created. Table 4 presents a comprehensive analysis of the deployed pre-trained models, emphasizing their accuracy achieved on the ImageNet dataset, total number of layers (depth) and the number of parameters.

4.4. Performance evaluation measures

This section provides an introduction to the performance factors used for the classification of prediction outcomes in the stage of Alzheimer's disease (AD). The proposed approach utilizes many evaluation metrics, including *Accuracy*, *Precision*, *Sensitivity*, *F1-score*, *AUC*, and *ROC curve*, to assess the reliability of the classification. These concepts are valuable in the field of machine learning and deep learning for assessing the predictive performance of the proposed model. The four outcomes often used to assess the effectiveness of classification systems are true positive (TP), true negative (TN), false positive (FP), and false negative (FN). These outcomes have considerable significance and are essential in evaluating the performance of such approaches. The performance criteria for the proposed techniques were derived from four distinct outcomes, which were afterwards used to evaluate the efficacy of these approaches. Furthermore, the feasibility and efficiency of the

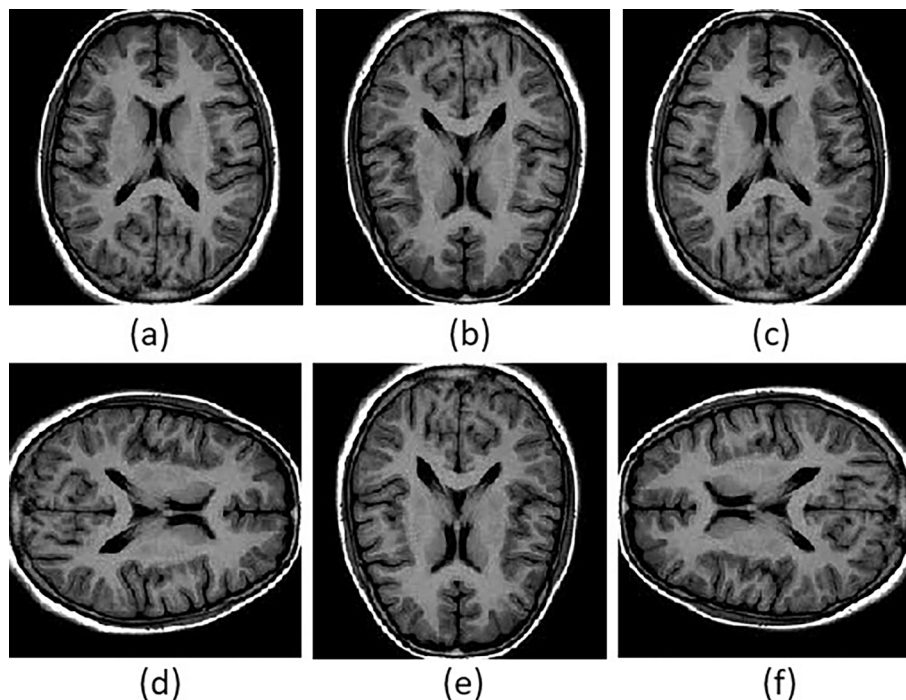


Fig. 4. Data augmentation applied on ADNI image (a) original image (b) vertical flip (c) horizontal flip (d)90° rotation (e) 180° rotation (f) 270° rotation.

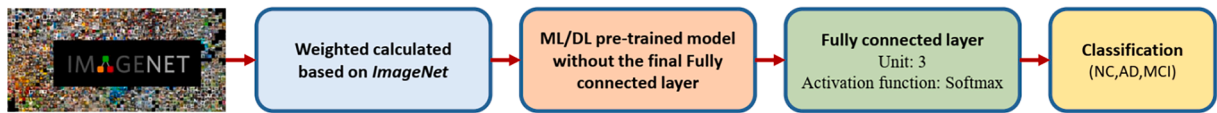


Fig. 5. Detailed description of the methodology used in constructing the baseline models using pre-trained deep neural networks.

Table 4

An analysis of the proposed model in relation to the pre-trained models that were employed.

Pre-trained model	Number of parameters	Number of Layers (Depth)	Accuracy achieved
ResNet152	60.2 M	152	84.2 %
Inception-v4	42.7 M	164	83.6 %
MobileNet-V2	3.5 M	53	78 %
Xception	22.9 M	126	81.3 %
GoogLeNet	6.7 M	22	77 %
SqueezeNet	1.2 M	18	82.7 %
VGGNet	143.7 M	19	85.7 %
AlexNet	61 M	8	78 %
DenseNet201	20.2 M	201	78.8 %
NaSNet-A	88.9 M	23	85.4 %
SENet	115.8 M	10–100	85.9 %
EfficientNet-B7	66 M	800	86.4 %
YOLO-V4	63 M	10–100	81.9 %
RetinaNet	64 M	10–100	84 %
Our Proposed model	1.9 M	18	92.3 %

methods were examined using the following indices. **Accuracy** refers to the fraction of data values that are correctly predicted in relation to the total accessible data. The formulation may be expressed as the ratio of the sum of true negatives and true positives to the sum of true negatives, true positives, false negatives, and false positives. **Precision** is the parameter that serves the purpose of discerning the affirmative prognostications generated by the model. The formula is expressed as the ratio of true positives to the total number of positive predictions. **Sensitivity** is the performance metric of a model may alternatively be expressed as the ratio of properly predicted positive values to the total number of positive samples. **F1-score** can be defined as the average value of sensitivity and precision which is used to measure the deep learning model's effectiveness. Insights may be gleaned from the dataset's uneven distribution of class instances. Also, the rates of false-positive and false-negative results need to be taken into account. **AUC** is the evaluation metric often referred to as the area under the ROC curve is used to evaluate the performance of a classifier. The AUC measure is used to assess the classification capability of a model in relation to the given dataset. A larger value for (AUC) score is indicative of a superior model. The AUC provides a measure of the effectiveness of using the analysis for predicting the outcome in a given situation. **ROC**: The term "receiver operating characteristic" is also often used to refer to this concept. The graphical representation is constructed with the purpose of visually demonstrating the effectiveness of classifiers in accurately identifying and diagnosing issues.

$$\text{Accuracy} = \frac{tp + tn}{tp + tn + fn + fp} \times 100\%$$

$$\text{Precision} = \frac{tp}{tp + fp} \times 100\%$$

$$\text{Sensitivity} = \frac{tp}{tp + fn} \times 100\%$$

$$\text{F1-score} = \frac{2tp}{2tp + fp + fn} \times 100\%$$

4.5. Performance result of the proposed model

The following section offers a brief overview of the performance results that were produced from the proposed model when it was applied to Alzheimer's disease Neuroimaging Initiative-1 (ADNI-1) dataset. Fig. 6 displays the outcomes of predictive analysis by visually representing the significant regions of interest (ROIs) in the brain for the diagnosis of Alzheimer's disease (AD) and mild cognitive impairment (MCI) using structural magnetic resonance imaging (sMRI) in our research study. Mild cognitive impairment (MCI) may represent an initial phase within the illness continuum, perhaps including the progression towards Alzheimer's disease (AD). Based on hyper-parameter optimization method, it was determined that the proposed model would need a cumulative number of 300 epochs for the purpose of training. The training process of the proposed model, however, persisted for a cumulative duration of 268 epochs until it was halted due to the implementation of the Early Stopping approach. According to the outcomes of the experiment, it has been determined that the optimal duration for the training period of the proposed model is 42.6 s. The Fig. 7 illustrates the accuracy values obtained during the training of the proposed model for both the training set and the validation set. The proposed model has been evaluated using six different parameters, which are as follows: accuracy, precision, sensitivity, F1-score, area under the curve (AUC), and ROC curve. The proposed approach demonstrates suitability for image classification, with the CNN serving as a superior feature extraction model. Subsequently, the LSTM and dense layer models exhibit enhanced performance in image classification. The combination of both models has shown superior results for the classification of Alzheimer's disease (AD). The findings indicate that the proposed model on ADNI-1 dataset has shown a level of accuracy of 92.3 % specifically pertains to the three-class classification of Alzheimer's disease (AD), Mild Cognitive Impairment (MCI), and Cognitively Normal (CN) subjects. Additionally, the precision, sensitivity, F1-score, and AUC values are reported as 92.1 %, 92.2 %, 92.25 %, and 97 %, respectively, as seen in Fig. 8. The Receiver Operating Characteristic (ROC) has also been used for the purpose of forecasting the model's performance, as seen in Fig. 9. The ROC curve illustrates the outcomes of the proposed model, indicating that the model exhibits superior performance by consistently appearing in the upper-left region of the graph.

Furthermore, the proposed model's ability is investigated in Table 5 to perform binary classification, particularly in distinguishing between Alzheimer's disease (AD) and Cognitively Normal (CN) as well as Mild Cognitive Impairment (MCI) and CN. An additional experiments were

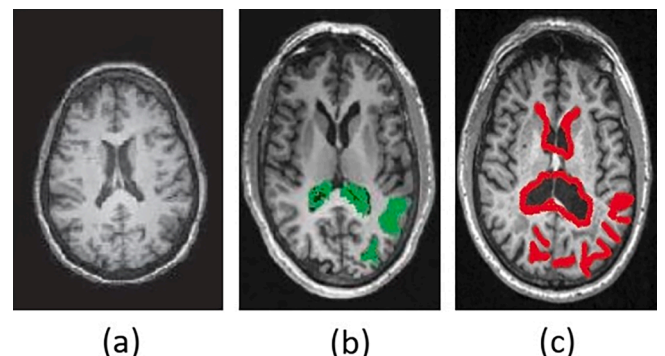


Fig. 6. Illustration of some predictive outcomes (a) NC (b) MCI (c) AD.

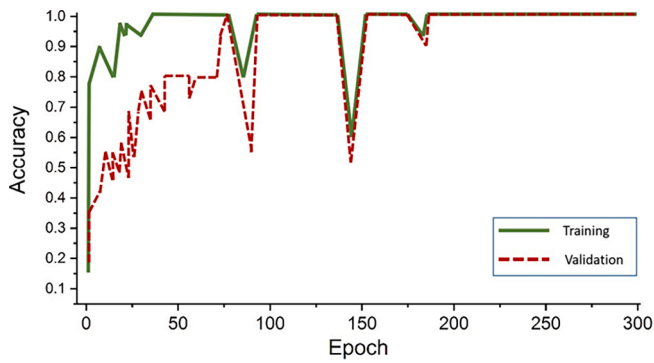


Fig. 7. Depiction of accuracy values attained for the training and validation sets are shown during the training process of the proposed model.

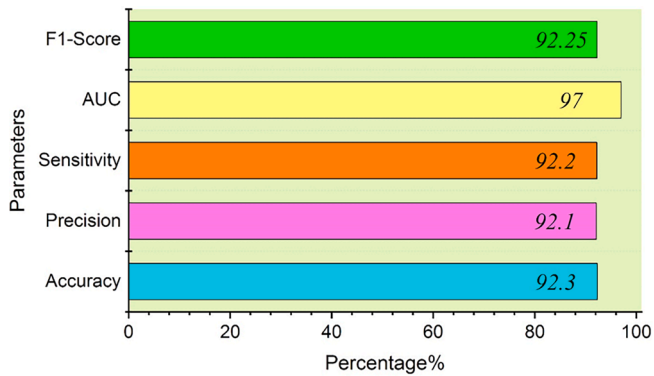


Fig. 8. Performance results of the proposed model on ADNI-1 dataset.

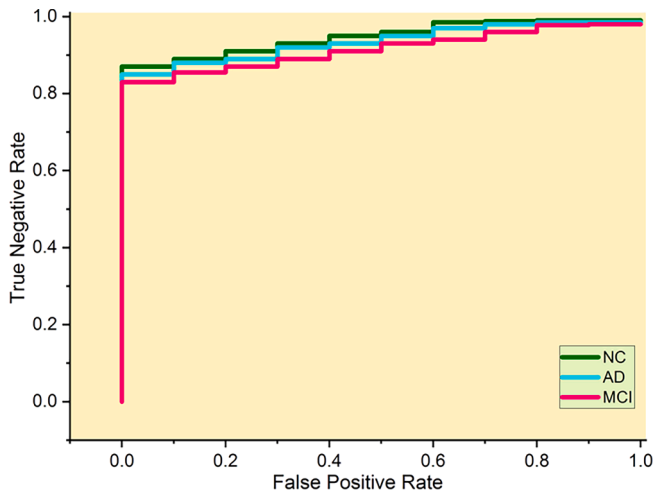


Fig. 9. ROC curve of the proposed model for NC, AD and MCI.

Table 5

Performances of the proposed model in binary classification task (%).

Classes	Accuracy	Precision	Sensitivity	F1-score
CN/AD	91.8	87.1	96.2	93.7
CN/MCI	92.5	86.2	93.3	90.1
MCI/AD	92.6	93.6	93.2	92.5

conducted to evaluate the model's performance in these binary classification tasks.

The plot of the attained accuracy values for the deployed machine

learning and deep learning models, as well as the proposed model, is shown in Fig. 10. It is undeniable that the model that was proposed had the greatest degree of accuracy, as it achieved an astounding 93.3 %. Following that, the NaSNet-A model and the Efficient-B7 model both produced results with an accuracy rate of 91.1 % and 91 % respectively, which was the second and third highest accuracy. By using the other ML, DL models and merging data from the tasks that were the most successful, were able to reach a comparable accuracy rate on ADNI-1 dataset. However, ML models show low accuracy rate compared to DL models which were deployed. Table 6 shows the performance comparison of different ML and DL models in terms of evaluation metrics.

The length of time needed to classify a particular input image is referred to as the inference duration, and it has a large substantial significance in the implementation of deep neural networks, particularly in real-time processing systems. The findings that are shown in Fig. 11 illustrate that the novel proposed model performed better than the deep neural networks (DNNs) in terms of inference time. The effectiveness of the system was shown by the fact that it was able to classify each test image in an average of only 1.7 ms. The aforementioned result reveals that the novel proposed model displays higher performance in comparison to employed DNN model, especially SqueezeNet and AlexNet, which is a well-known deep neural network architecture that is recognized for its lightweight design. SqueezeNet and AlexNet is well-known in the research community for having an architectural design that is both effective and lightweight. The empirical data that was acquired provides evidence for the prospective usefulness of the unique technique that was presented, particularly when used to the context of lightweight or real-time systems. In contrast, it was noted that EfficientNet-B7, DenseNet201 and Inception-v4 had the longest inference time out of all the models that were investigated, which rendered it unsuitable for usage in a real-time system when it comes to classification of AD.

Despite being executed on powerful computers with high processing capacity, such as advanced GPUs or TPUs, and sufficient memory, the training of deep neural networks is well recognized to be a time-consuming procedure. The complex and profound design of the neural network framework, the incorporation of a large number of parameters, and the processing of significant volumes of data are the three key factors which attributed to be the primary reasons for this phenomenon. Hence, we illuminate the duration required to train each used deep neural network, as shown in Fig. 12, inside our experimental analysis. Based on the experimental findings, it was determined that SqueezeNet exhibited the highest speed among the models tested, finishing the training process in a time frame of 29.5 s. The training process for AlexNet and the proposed model followed after SqueezeNet, with completion times of 30.4 and 32.6 s, respectively. The EfficientNet-B7 model was identified as the slowest among the deep neural networks used, which is to be expected given its significant depth of 800 layers. It can be inferred from the results of this experimental investigation that there is a negative association between the depth of a neural network and the amount of time needed for training. This suggests that shallower networks have a tendency to have shorter training durations all around. The amount of storage space needed to hold the weights exhibits a direct relationship with the quantity of weights, whereas the computational effort necessary in performing the convolution operation demonstrates a linear correlation with the number of operations. The degree to which an algorithm is complicated may be represented mathematically as a function of the input data that it receives. It is possible to determine the level of computational complexity posed by a Convolutional Neural Network (CNN) by adding up the levels of complexity posed by its individual layers. Filter size, size of input and filters number are the three key factors that determined the computational complexity of a convolutional layer.

4.6. The efficiency of the loss function

The assessment of the focal Tversky loss (FTL) using the ADNI-1

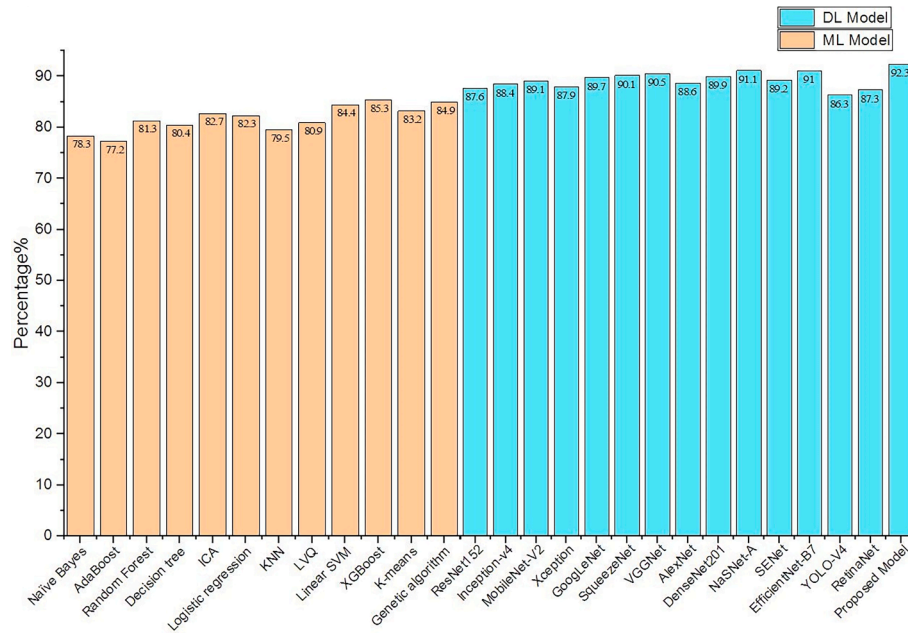


Fig. 10. The depicted plot shows accuracy values of the models that were employed to represent how well they performed while being evaluated on the test set. The model that was proposed was successful in achieving the greatest degree of accuracy, which was 92.3%.

Table 6

The performances comparison of employed models based on evaluation metrics (%).

Method	Accuracy	Precision	Sensitivity	F1-score
Naïve Bayes	78.3	74.1	88.2	83.7
AdaBoost	77.2	73.2	84.3	77.1
Random Forest	81.3	78.8	94.7	81.2
Decision tree	80.4	76.5	84.1	80.4
ICA	82.7	80.2	86.7	82.5
Logistic regression	82.3	79.3	85.4	81.8
kNN	79.5	93.8	65.4	71.3
LVQ	80.9	87.4	82.3	79.1
Linear SVM	84.4	76.4	90.1	85.0
XGBoost	85.3	77.3	93.2	86.4
K-means	83.2	79.4	88.8	80.3
Genetic algorithm	84.9	79.3	87.2	84.7
ResNet152	87.6	89.3	87.3	87.7
Inception-v4	88.4	90.2	88.2	88.5
MobileNet-V2	89.1	92.1	88.8	89.1
Xception	87.9	89.7	87.3	87.8
GoogLeNet	89.7	90.3	88.3	89.6
SqueezeNet	90.1	90.9	89.7	90.2
VGGNet	90.5	90.9	90.3	90.6
AlexNet	88.6	89.8	87.5	88.3
DenseNet201	89.9	90.0	89.8	89.9
NaSNet-A	91.1	90.2	90.8	91.2
SENet	89.2	87.3	89.0	89.3
EfficientNet-B7	91	90.2	91.2	91.1
YOLO-V4	86.3	84.3	85.8	86.1
RetinaNet	87.3	86.9	89.5	87.5
Proposed Method	92.3	92.1	92.2	92.25

dataset, as shown in Fig. 13, illustrates its efficacy in identifying Alzheimer's disease. By outperforming all other methods in terms of Dice coefficient, the trials demonstrated that the FTL methodology excels in segmenting and localizing important brain areas in MRI images. The improved performance is due to a mix of Binary Cross-Entropy Loss (BCE-Loss) and Dice-Loss, which find a happy medium between the requirements for sensitive and precise detection of Alzheimer's disease. Cross-Entropy Loss (CE-Loss) mediates the efficacy of the FTL approach and the Dice-CE loss, according to the results. This means that it helps to decrease class imbalance issues without reducing the efficiency of FTL.

In contrast, using Dice-Loss alone results in the lowest Dice coefficient, suggesting that this method may struggle to deal with complicated data and provide inaccurate predictions. Based on these findings, it is reasonable to assume that the FTL approach reduces false negatives and effectively identifies difficult samples, thereby increasing the accuracy of Alzheimer's disease diagnoses.

4.7. Implementation details via software stack

The PyTorch [69] module was used in all experiments, which were executed on Python 3. The model was trained on a Windows x86-64 workstation equipped with an Intel(R) Xeon(R) processor operating at a frequency of 3.60 GHz, an NVIDIA Quadro P620 graphics card, and 32 GB of random access memory (RAM), for a total of 100 iterations. During the process of constructing this module, a variety of modern open-source Python libraries were used. PyTorch module is a high-level application programming interface (API) that enables the implementation of deep neural networks on a variety of deep learning backends, such as TensorFlow [70] and Theano. Both the proposed approach and the baseline model were designed using deep neural networks and then implemented using PyTorch module. The proposed model uses a Google-owned, cutting-edge deep learning backend after it was recommended by pytorch's developer and became the default option. An esteemed Python package scikit-learn [71] was used to use widely-used conventional machine learning models as the benchmarks for the proposed model and to execute a number of operations on the dataset, including cross-validation and subset partitioning. Scikit-learn incorporates several traditional machine learning methods, albeit it lacks implementations for certain baseline models used in the proposed methodology. Therefore, the scikit-learning vector quantization (sklqv) [72], LightGBM [73] and XGBoost [74] (Python) packages served as the basis for the implementation of these approaches. Large, multidimensional arrays and matrices can readily operate by NumPy [75]. The pandas [76] library, was used for the purpose of manipulating numerical tables. Matplotlib [77] and seaborn [78] were used to plot the experimental results of the performed procedures.

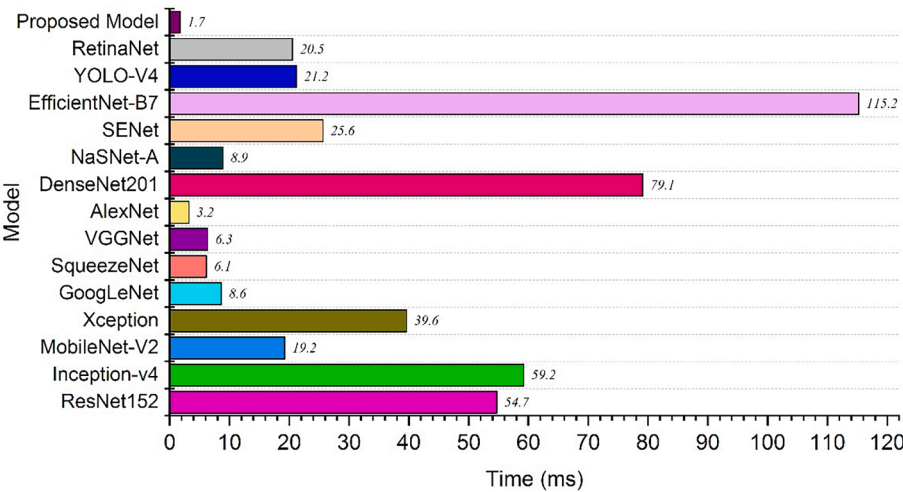


Fig. 11. The plot depicts the inference time for each deployed deep neural network (DNN).

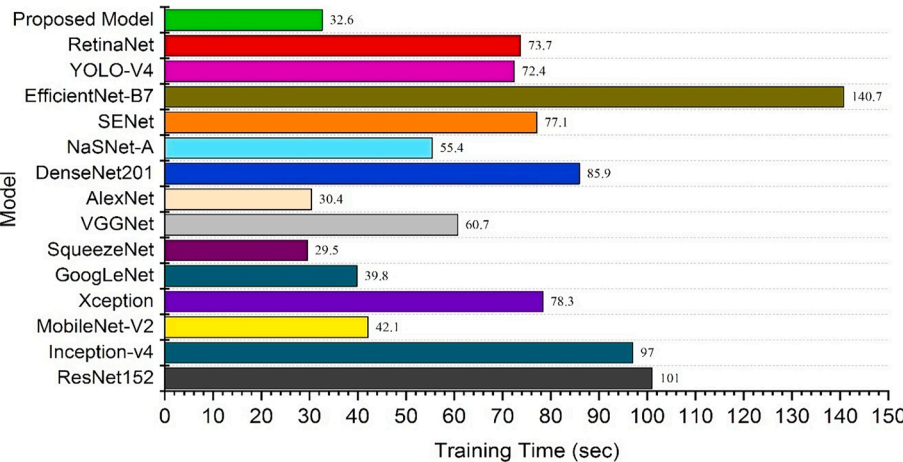


Fig. 12. Depiction of time required to train each DNN.

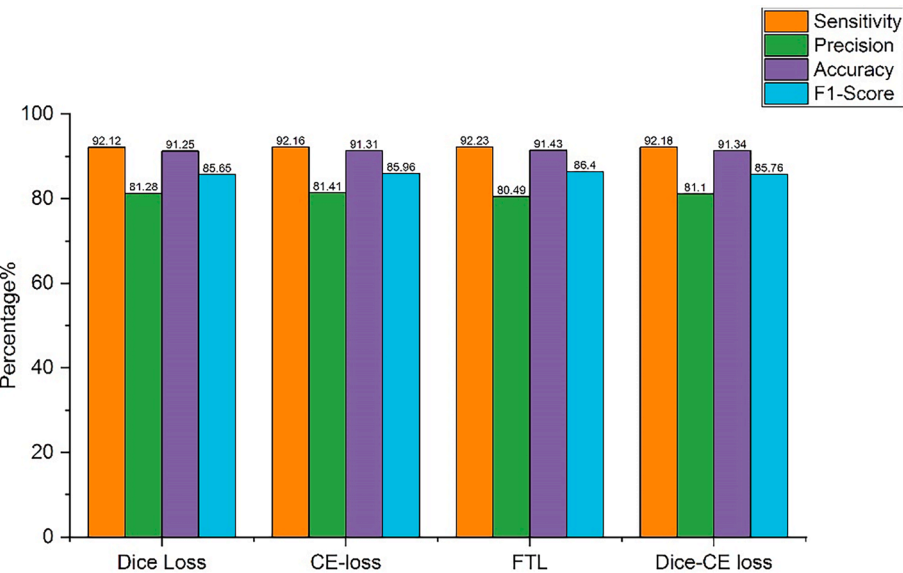


Fig. 13. Performance comparison of loss function.

4.8. Comparative analysis with other state-of-the-art techniques

We contrasted the performance of our proposed model with that of other prominent models to emphasize its advantages. A comparative assessment of the proposed model's classification accuracy against state-of-the-art models is shown in Table 6. The table comparison clearly indicates that we have achieved a superior accuracy rate relative to prior research. Previous study used deep learning and other notable models to address multistage Alzheimer's disease predictions, namely NC, AD, and MCI, using the same ADNI dataset. Table 7 illustrates the graphical comparison of the proposed model to other dominant models.

4.9. Ablation study

The dataset used in this research was subjected to a series of ablation experiments to determine how different structural configurations might affect the proposed architectural framework. The ablation study determines how well our proposed method works by analyzing the ADNI dataset.

4.9.1. Impact of data augmentation

Convolutional neural networks (CNNs) often use data augmentation, particularly when datasets are limited. Our study of endoscopic images revealed several instances of both healthy and malignant tissues. To enhance the representativeness of the samples relative to the broader population, augmentation techniques such as translation, flipping, and rotation are used. Flip augmentation involves horizontally and vertically mirroring images, while rotation—specifically in increments of 90°—enhances dataset size. This research examines the impact of various strategies on critical performance metrics. The metrics in Table 8 indicate that data augmentation significantly improves the performance of the proposed model.

4.9.2. Impact of input resolution

The model's effectiveness varies with the input data size, as seen in Fig. 14, which presents dimensions of 224×224 and 448×448 . Alterations in resolution provide varying degrees of image scaling, hence affecting the model's outcomes. The computational complexity increased significantly, although the accuracy showed little improvement despite enhanced resolution.

4.9.3. Influence of convolutional layers count

The performance of a model is significantly affected by the number of

Table 7
Comparative analysis with the state-of-the-art techniques (%).

Study	Model	Dataset	Accuracy
A.Khan et al. [98]	CNN	ADNI	87.2 %
S. Grueso et al. [99]	CNN	ADNI	78.5 %
Katabathula et al. [100]	3D DenseCNN	ADNI	92.2 %
Hu et al. [101]	Pretrain model + Transformer	ADNI	77.2 %
Hazarika et al. [102]	Hybrid pretrained models	ADNI	88 %
Zhang et al. [103]	Multi-model Deep learning	ADNI	88.5 %
Liu et al. [104]	Multi-modal Deep CNN	ADNI	88.9 %
T.Zhang et al. [105]	deep multi-modal fusion network	ADNI	86.1 %
Qiao et al. [106]	Convolutional- Squeeze-Excitation- Softmax-NET (CSES-NET)	ADNI	72.4 %
S.Q.Abbas et al. [107]	3D Jacobian domain CNN	ADNI	91.2 %
Y.Tong et al. [108]	Spatial context 3D-network	ADNI	81.2 %
Our Proposed	Lightweight CNN-LSTM model	ADNI	92.3 %

Table 8

Ablation study on the influence of data augmentation ([In %]).

Data augmentation	Classes	Accuracy	Precision	Sensitivity	F1-score
Without data augmentation	CN/AD	89.7	85.3	94.1	91.5
	CN/MCI	90.8	84.2	91.6	88.6
	MCI/AD	90.4	91.7	91.8	90.6
	CN/AD	91.8	87.1	96.2	93.7
With data augmentation	CN/MCI	92.5	86.2	93.3	90.1
	MCI/AD	92.6	93.6	93.2	92.5
	CN/AD				
	MCI/AD				

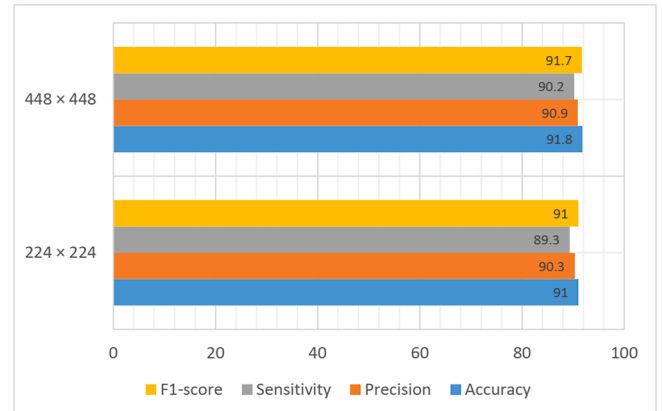


Fig. 14. Performance of the proposed model in different input resolutions.

convolutional layers. An excessive number of layers will result in elevated processing costs, while an insufficient number would compromise the accuracy of contextual data extraction. The proposed model may have one to three convolutional layers, and we examined the effects of varying that quantity. The addition of layers improved performance, as seen in Table 9. The model's ability to extract global information is enhanced by the inclusion of convolutional layers before to the input, hence increasing its overall effectiveness.

4.9.4. Impact of LSTM layer

Fig. 15 illustrates the model's classification performance by comparing the CNN with and without LSTM components. Utilizing LSTM enhanced the model's comprehension of the data's temporal correlations, hence improving its accuracy and robustness in classification tasks. The CNN-LSTM model exhibited superior classification accuracy compared to the CNN-only configuration, indicating that the integration of CNN's effective feature extraction with LSTM's sequential processing produces enhanced outcomes. Utilizing both methodologies is essential for optimal results in complex data situations.

4.9.5. Impact of 3D data input

Fig. 16 displays the model's classification performance when fed 3D data. The findings showed that the model worked better using 2D data, even though 3D data was originally planned to be used. The lack of enhanced accuracy or resilience in classification tests suggests that there

Table 9

Ablation study on the number of convolutional layers ([In %]).

Convolutional layers	Accuracy	Precision	Sensitivity	F1-score
1	91.8	90.3	89.1	91.4
2	92.0	90.6	89.5	91.7
3	92.3	90.8	89.8	91.9

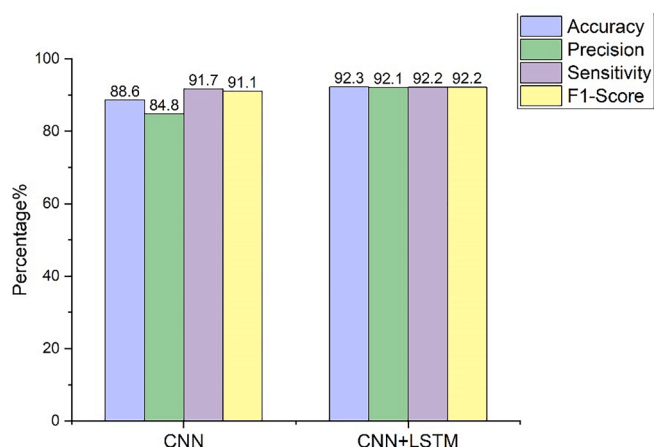


Fig. 15. Performance of the proposed model with and without LSTM layer.

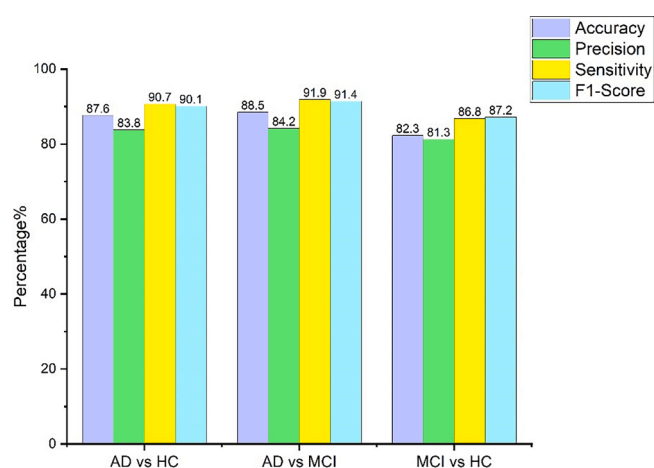


Fig. 16. Performance of the proposed model in 3D data input.

may be issues in properly collecting essential characteristics, even though the 3D dataset is rich in spatial information. It is essential to provide these results alongside the 2D ones since it shows how the 3D method doesn't work here and how well the 2D format works for our particular categorization goals.

5. Conclusion

Alzheimer's disease (AD) is one of the most debilitating and sometimes fatal neurological disorders. The current state of knowledge on this disease's therapy is unfortunate. The progressive nature of Alzheimer's disease (AD) symptoms makes it more difficult to go about one's everyday life. When it comes to medical AD classification, deep learning approaches, especially Convolutional Neural Networks (CNN), have been much more successful than conventional machine learning techniques. Specifically, it may be used without the requirement for any kind of laborious feature extraction. There will likely be a dramatic rise in the incidence of neurodegenerative diseases in the next decades as a result of the worldwide trend of increased longevity among individuals with Alzheimer's disease (AD) and dementia. This research presents an innovative lightweight CNN-LSTM based method for classifying AD, MCI, and NC, which is quick, accurate, and inexpensive approach. We initially utilized a widely accepted and reliable dataset called ADNI-1, and was developed to differentiate between AD, MCI, and NC. The 505 T1-weighted 3D structural MRI images that made up this dataset were collected from various participants including total of 135 people with AD, 215 with MCI, and 155 with NC. Our best understanding is that

the Alzheimer's disease Neuroimaging Initiative (ADNI-1) is the largest publicly available dataset for the diagnosis of AD, MCI, and NC which comprises three-dimensional features. Afterwards, the transfer learning method was used to convert the three-dimensional features into two-dimensional features. The newly developed dataset was used to train and evaluate the novel model. In the final phase FTL is used as a loss function to extract challenging data and reduce the number of false-negative predictions. The experimental results show that the proposed approach attained an acceptable 92.3 % accuracy. To evaluate the new model's performance, a collection of 21 state-of-the-art classical ML methods and DNNs were utilized as comparison points. In comparison to the baselines, the proposed approach attained a higher accuracy of 92.3 %. The proposed innovative model outperformed both deep neural networks (DNNs) and state-of-the-art conventional machine learning methods with an average inference time of 1.7 ms. A lightweight or real-time system might benefit from the proposed approach, as shown by these experimental results. In our future research, we aim to utilize several strategies during the conversion of features from 3D to 2D in order to assess the impact of these techniques on the accuracy of the proposed model. We aim to enhance the learning capacity of the proposed innovative model by training it on a more extensive multi-modal dataset. Furthermore, we will investigate the actual implementation of the model and further develop this approach for detecting other types of cancers.

CRedit authorship contribution statement

Ejaz Ul Haq: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Qin Yong:** Supervision, Project administration. **Zhou Yuan:** Software, Project administration. **Xu Huarong:** Supervision. **Rizwan Ul Haq:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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